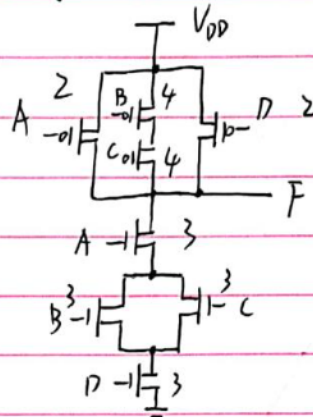


《数字 VLSI》第二次作业

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$$\begin{aligned} 1. \quad F &= \bar{A} + \bar{B}\bar{C} + \bar{D} \\ \bar{F} &= \overline{\bar{A} + \bar{B}\bar{C} + \bar{D}} \\ &= \bar{\bar{A}} \cdot \bar{\bar{B}\bar{C}} \cdot \bar{\bar{D}} \\ &= A \cdot (B+C) \cdot D \end{aligned}$$



2. (1) 上升传播延时 t_{PLH} , 此时上管导通, 下管截止

① 初: $V_{G1} = V_{GS} - V_{TH} = 2.07V$, $V_{DS} = V_{DD} - 0 = 0.5V$, $V_{DSAT} = 0.63V$

$$\therefore I = k' \cdot \frac{W}{L} (V_{G1} \cdot V_{DS} - \frac{1}{2} V_{DS}^2) = 418.6 \mu A$$

$$R_1 = V_{DS} / I = 1.194 k\Omega$$

② 末: $V_{G1} = V_{GS} - V_{TH} = 1.82V$, $V_{DS} = V_{DD} - \frac{1}{2} V_{DD} = 0.25V$, $V_{DSAT} = 0.63V$

$$\therefore I = k' \cdot \frac{W}{L} (V_{G1} \cdot V_{DS} - \frac{1}{2} V_{DS}^2) = 194.9 \mu A$$

$$R_2 = V_{DS} / I = 1.283 k\Omega$$

③ ~~$R_{eq1} = \frac{1}{2} (R_1 + R_2)$~~ $t_{PLH} = 0.69 (R_1 + R_2) / 2 \cdot C = 855 ps$

(2) 下降传播延时 t_{PHL} , 此时下管导通, 上管截止

① 初: $V_{G1} = V_{GS} - V_{TH} = 2.07V$, $V_{DS} = 0.5V$, $V_{DSAT} = 0.63V$

$$\therefore I = k' \cdot \frac{W}{L} (V_{G1} \cdot V_{DS} - \frac{1}{2} V_{DS}^2) = 418.6 \mu A$$

$$R_1 = V_{DS} / I = 1.194 k\Omega$$

② 末: $V_{G1} = V_{GS} - V_{TH} = 2.07V$, $V_{DS} = 0.25V$, $V_{DSAT} = 0.63V$

$$\therefore I = k' \cdot \frac{W}{L} (V_{G1} \cdot V_{DS} - \frac{1}{2} V_{DS}^2) = 223.675 \mu A$$

$$R_2 = V_{DS} / I = 1.118 k\Omega$$

③ $t_{PHL} = 0.69 (R_1 + R_2) / 2 \cdot C = 798 ps$

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3. $t = 0.69 [C_1 R_5 + C_2 R_5 + C_3 R_5 + C_4 R_5 + C_5 \cdot 0 + C_6 (R_5 + R_6) + C_7 R_5 + C_8 R_5 + C_9 (R_5 + R_6 + R_7)] = 759 ps$